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Entropy generation analysis of Carreau-Yasuda hybrid nanofluid flow with Thompson-Troian boundary conditions and Cattaneo-Christov heat flux

S. Suneetha^a, Rajavath Narayana Naik^a, K.S. Srinivasa Babu^b and M. Jayachandra Babu^c

^aDepartment of Applied Mathematics, Yogi Vemana University, Kadapa, India; ^bDepartment of EM&H, S.R.K.R. Engineering College, Bhimavaram, India; ^cDepartment of Mathematics, Government Degree College, Rajampeta, India

ABSTRACT

This work scrutinises the flow and heat transmission features of a hybrid nanofluid with Carreau-Yasuda distribution as it flows across an extending sheet with the Thompson-Troian boundary conditions. The impacts of different parameters, including the Weissenberg number, thermal relaxation parameter on the velocity, temperature and concentration profiles are presented. After being converted to a set of non-linear ordinary differential equations, the governing equations are numerically work out with bvp4c method in MATLAB. The friction coefficient exhibits a 34.4% increase as the Carreau-Yasuda value rises from 0 to 0.5, hence emphasising the influence of shear-thinning mechanisms on fluid flow. The thermal relaxation parameter exerts a more significant influence on the Nusselt number than on the Eckert number in the context of heat transport. Enhancing the thermal relaxation parameter from 0 to 0.4 results in an 8.8% hike in Nusselt number, indicating improved heat transfer efficiency. The Sherwood number exhibits a reduction of 3.2% when the activation energy parameter rises from 0 to 3. The Brinkman number and the volume percentage of nanoparticles have substantial impacts on entropy generation as well as the Bejan number. Specifically, higher values of these parameters result in more entropy creation and a lower Bejan number.

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KEYWORDS

Non-Newtonian fluid; viscous dissipation; non-Fourier heat flux; bvp4c; activation energy

Nomenclature

u, v	Velocity components in x, y directions respectively [m/s]
J	Diffusion parameter
ρ	Fluid density [kg/m^3]
μ	Dynamic viscosity [$kg/(m.s)$]
Br	Brinkman number
β_1	Concentration difference parameter
ν_w	Permeability of porous surface [m^2]
σ	Electric conductivity [S/m]
T	Fluid temperature [K]
ζ^*	Critical shear rate [$1/s$]
Υ	Relaxation time of the fluid [s]
C_p	Specific heat capacity [$J/kg \cdot K$]
k_0	Chemical reaction parameter [$1/s$]
B_0	Magnetic field strength [T]
γ^*	Navier's slip length [m]
ν	Kinematic viscosity [m^2/s]
C	Fluid concentration [mol/m^3]
k^*	Mean absorption coefficient [$1/m$]
σ^*	Stefan–Boltzmann constant
D_m	Molecular diffusivity [m^2/s]
Λ	Thermal relaxation parameter
Sc	Schmidt number
We	Weissenberg number
Ra	Radiation parameter
Eck	Eckert number
Bn	Bejan number

Br	Brinkman number
θ	Fluid temperature (non-dimensional)
Φ	Fluid concentration (non-dimensional)
EG	Entropy generation (non-dimensional)
Mg	Magnetic field parameter

1. Introduction

Nanofluids are built with ultrafine particles (≤ 100 nm) suspended in the variety of conventional fluids, including oil, ethylene glycol, and medical fluid oil. The heat-transmission properties of the fundamental fluids can be enhanced by using nanoparticles. Widely used metal and non-metal nanoparticles improve carrier fluids' thermal characteristics dramatically. Nanofluids can be used in defence systems, refrigeration, solar heating, power hybrid plants, and many other applications. An introductory attempt on these fluids was made by Choi (2009). A report on nanofluids by Saidur, Leong, and Mohammed (2011) through which they discussed the characteristics and convection phenomenon of nanofluids. Philip and Shima (2012) presented the thermal properties of nanofluids. Through this research, it was found that during measurements, even slight temperature differences in the sample might cause significant inaccuracies in the thermal conductivity readings. The effective use of nanofluids and their preparations are studied by Sidik et al. (2014). An extensive discussion of the techniques for preparing nanofluids is presented in this work. Despite the numerous preparation techniques that have been suggested,

creating a homogenous, long-term stable nanofluid that doesn't agglomerate much and doesn't change its thermophysical characteristics remains a difficulty. Venkateswarlu, Varma, and Prasad (2022) have reported the numerical findings of a nanofluid movement past an extending surface under convective boundary conditions along with the non-Fourier heat flux effect. Revathi et al. (2023a, 2023b) numerically studied the movement of Ostwald-de Waele type nanofluid with thermal radiative effects. The convective motion of a non-Newtonian nanofluid over an extending surface under the impact of thermal radiation is another study by Poornima, Sreenivasulu, and Souayah (2023). Over the years, the usage of nanofluids in the diversified carrying fluids has mainly focused on several discussions and different mathematical investigations. The more relevant studies on these fluids are based on the mixing up of the variety of two or more nanoparticles (hybrid nanofluid models) into the base fluids are found in literature from the last few years. Extensive work has been conducted by several researchers to see the impacts on the thermal characteristics of the fluids due to the involvement of a variety of nanoparticles. Sarkar, Ghosh, and Adil (2015) deliberated the behaviour of hybrid nanofluids. This article reviews the synthesis, thermophysical features, heat transmission, and features of pressure drop of several hybrid nanofluids based on experimental as well as numerical studies conducted by various reporters. Al Oweidi et al. (2022) scrutinised the modelling of a ternary hybrid nanofluid along a rotating disk with thermal radiation using the Keller Box scheme. Afterwards, subsequent studies (Nazir et al. 2022; Anantha Kumar et al. 2023; Khan et al. 2023; Nazeer et al. 2023; Nazir et al. 2023; Revathi et al. 2023a, 2023b; Madhu et al. 2024; Hayat et al. 2024; Venkateswarlu, Narayana, and Joo 2024; Kumar 2024; Zeeshan et al. 2024a, 2024b; Mehboob, Ellahi, and Sait 2024; Nazir et al. 2024; Irfan et al. 2024) investigated a range of nano/hybrid nanofluid flows across distinct geometries based on the influence of various parameters, including thermal radiation.

In many relevant circumstances, the most effective model for describing the heat transmission mechanism has been the well known Fourier's heat conduction law (1888). Its primary drawback, nonetheless, is that it produces a parabolic energy equation for the temperature domain, which runs counter to the causality property. By adding a crucial component of thermal relaxation time, modification of the Fourier's model prescribed by Cattaneo. Heat can be transported by thermal waves propagating at a finite speed according to this concept, which yields a hyperbolic energy equation for the domain of temperature. Based on this fact, a huge number of research articles reported. Han et al. (2014) discussed the coupled flow, transfer of heat of a viscous fluid under the control of the Cattaneo-Christov heat flux model based on the homotopy analysis method. The characteristics of magnetohydrodynamic fluid motion through two different geometries, a wedge and a cone, were examined by Kumar et al. (2018) in their discussion of the non-Fourier heat flux model. It turns out that the Nusselt number spikes when the thermal relaxation parameter goes up. Kumar, Sugunamma, and Sandeep (2019a) discussed the movement of micropolar fluid with the heat source using the shooting technique. Kumar et al. (2019b) analysed the movement of two non-Newtonian fluids through an elastic cylinder using the Cattaneo-Christov heat flux model. Anantha Kumar, Sugunamma, and Sandeep

(2020) explored the movement of the Casson fluid through an elastic sheet with various parameters, including varying thermal conductivity. Mallawi et al. (2021, 2022) explored distinct non-Newtonian fluid movements across different plate geometries using the non-Fourier heat flux model. Based on this model, Eswaramoorthi et al. (2021; Eswaramoorthi and Sivasankaran 2022) scrutinised the Williamson fluid movement across an elastic sheet with chemical reaction parameters. Khattak et al. (2022) presented the numerical results based on the same model for a flow through a porous medium. Three-dimensional nanofluid modelling has been studied by Gowda et al. (2023) through the modified heat flux model. Shah, Rooman, and Shutaywi (2023), Noreen et al. (2023), and Farooq et al. (2023) are the researchers who reported the results of the various fluid motions through the Cattaneo-Christov heat flux (CCHF) modelling. A theoretical report to analyse the nanofluid motion over a stretchable cylinder is addressed by Ashraf, Tanveer, and Ulhaq (2023). Later, several studies (Loganathan et al. 2023; Eid et al. 2024; Narayana Naik et al. 2024; Hussain and Mao 2024; Eswaramoorthi et al. 2024) examined a variety of fluid flows over different geometries under the control of Cattaneo-Christov heat flux model.

Besides the variety of physical impacts on the flow motion, the activation energy impact also carries considerable significance for diversified numerical investigations. Significant results are visualised based on the activation energy in the fluid flow models. For instance, on the rotating disk, the nanofluid flow study under activation energy is another investigation of Hamid and Khan (2018) in which they showed how nanofluid concentration is affected by activation energy. The examination of activation energy as well as chemical reaction in the nanofluid flow based on a numerical procedure is addressed by Khan, Kumam, and Thounthong (2020). They presented that there is a decrement in concentration caused by the activation energy. Imparting the effect of activation energy in the nanofluid flow, Gowda et al. (2023) depicted the numerical outcome for the nanofluid's concentration profiles affected by activation energy impact. Recently, several investigators (Gautam et al. 2024; Awais, Salahuddin, and Muhammad 2024; Jabeen et al. 2024; Ahmed et al. 2024; Patil and Benawadi 2024; Madhu et al. 2024) discussed different fluid movements subject to various physical impacts, including activation energy.

Non-Newtonian fluids have intricate and intriguing flow behaviours that defy the straightforward linear connection between shear stress and shear rate that controls Newtonian fluids. Their viscosity changes in response to applied stress, leading to diverse applications across industries. In food processing, they enhance texture and stability, while in pharmaceuticals and cosmetics, they control release rates and improve product consistency. Understanding and modelling non-Newtonian fluid flows is crucial for optimising industrial processes, designing efficient pipelines, and even unravelling the mysteries of biological systems like blood flow. In recent times, Rehman et al. (2024a, 2024b, 2024c, 2024d, 2024e) examined diverse non-Newtonian fluid dynamics across flat surfaces employing neural network methodology.

The insights gained from this analysis of Carreau-Yasuda (C-Y) hybrid nanofluid flow with non-Fourier heat flux have significant real-world applications. Industries such as microelectronics,

where efficient heat dissipation is crucial, can leverage the findings to optimise cooling systems. The analysis of non-Newtonian fluid behaviour can be applied in food processing, pharmaceuticals, and biomedicine, where materials often exhibit non-Newtonian properties. Additionally, the study of activation energy effects can be beneficial in chemical engineering processes, particularly those involving catalytic reactions. Researchers and engineers may create more effective and efficient systems for a variety of applications by comprehending how different factors affect fluid flow and heat transfer. The novelty of this study lies in its comprehensive investigation of the Carreau-Yasuda hybrid nanofluid flow under the combined impact of viscous dissipation, thermal radiation, activation energy, and Cattaneo-Christov heat flux. While previous studies have explored some of these effects individually, this study provides a more holistic understanding of their interactions and their impact on the flow and heat transfer features. Additionally, the use of Thompson-Troian boundary conditions to account for slip and jump effects at the solid-fluid interface is a significant contribution to this research. This introduces additional complexity to the problem but provides a more accurate representation of the physical phenomena involved.

The following questions are the focus of this study:

- What effects does the fluid flow have when the Weissenberg number varies?
- What effects does the Cattaneo-Christov heat flux model have on the rate of heat transfer and temperature distribution?
- How does the inclusion of activation energy influence the concentration profile and rate of mass transfer?
- How does the variation in the volume fraction of nanoparticles affect the fluid flow, Bejan number and entropy generation?

2. Formulation

The Carreau-Yasuda hybrid nanofluid flow across an elastic sheet under Thompson-Troian boundary conditions is theoretically analysed in this report. In this investigation, we presuppose the following:

- The flow is assumed to be incompressible, steady, two-dimensional and laminar.
- The flow is analysed under the boundary layer approximation, where the flow is assumed to be confined to a thin region near the surface.
- A vertically imposed external magnetic field of strength B_0 affects the flow (see Figure 1).
- Viscous dissipation, non-Fourier heat flux, and thermal radiation parameters were considered in the energy equation to talk over the characteristics of heat transmission.
- Kindly see Table 1 for exact numerical calculations of the various attributes related to $WEG(H_2O + EG)$, Cu and TiO_2 such as density.
- The induced magnetic field due to the fluid motion is omitted in the presence of applied magnetic field.

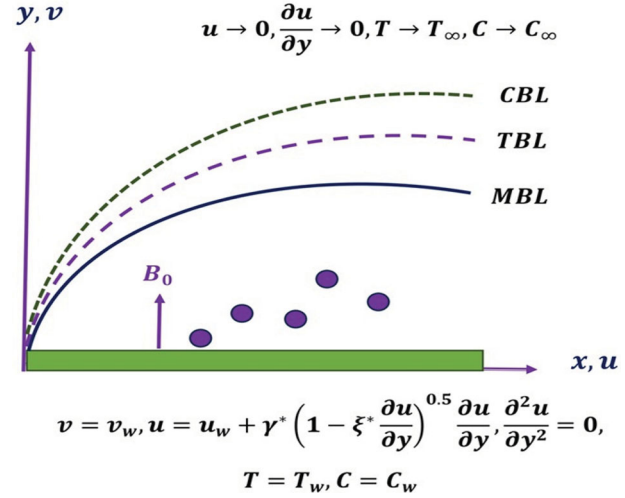


Figure 1. Visual depiction of the present formulation.

Table 1. Key parameters' values of $WEG(H_2O + EG)$, TiO_2 and Cu (Das et al. 2022).

S. No.	Properties	$WEG(H_2O + EG)(f)$	$Cu(\phi_C)$	$TiO_2(\phi_T)$
1	σ	2.06×10^{-6}	59.6×10^6	2.6×10^6
2	k	0.1816	401	8.9538
3	ρ	1041.89	8933	4250
4	C_p	3421.54	85	686.2

- The flow is driven solely by the stretching of the sheet, and there is no external pressure gradient.

In light of the above assumptions, vector notation of the equations describing the flow, i.e. continuity, momentum (without Carreau-Yasuda and convection terms), energy (without Cattaneo-Christov heat flux term), and concentration are presented below:

$$\nabla \cdot V = 0, \quad (1)$$

$$V \cdot \nabla V = \nu_{hnf} \nabla^2 V - \frac{\gamma_0}{\rho_{hnf}} \nabla^4 V - \frac{1}{\rho_{hnf}} J \times B, \quad (2)$$

$$(V \cdot \nabla T) = \frac{1}{(\rho C_p)_{hnf}} (\nabla \cdot k_{hnf} \nabla T) - \frac{1}{(\rho C_p)_{hnf}} (\nabla \cdot (qr)) + \frac{\mu_{hnf}}{(\rho C_p)_{hnf}} \Theta, \quad (3)$$

$$V \cdot \nabla C = D_m \nabla^2 C - k_0^2 \left(\frac{T}{T_\infty} \right)^n \exp \left(-\frac{E_0}{k_1 T} \right) (C - C_\infty) \quad (4)$$

See the following for the regular form of the equations along with the conditions (Khan et al. 2018; Irfan et al. 2019; Rao, Thiagarajan, and Kumar 2022 and Nadeem, Ahmad, and Khan 2021):

$$\frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} = 0 \quad (5)$$

$$\begin{aligned} v \frac{\partial u}{\partial y} + u \frac{\partial u}{\partial x} &= \frac{\mu_{hnf}}{\rho_{hnf}} \frac{\partial^2 u}{\partial y^2} + \frac{\mu_{hnf}}{\rho_{hnf}} \frac{\partial^2 u}{\partial y^2} \Upsilon^r \left(-\frac{m-1}{r} \right) (r+1) \left(\frac{\partial u}{\partial y} \right)^r \\ &+ \beta_c (C - C_\infty) g - \frac{\gamma_0}{\rho_{hnf}} \frac{\partial^4 u}{\partial y^4} + \beta_T (T - T_\infty) g \\ &- \rho_{hnf}^{-1} \sigma_{hnf} u B_0^2 \end{aligned} \quad (6)$$

$$\begin{aligned}
u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} &= k_{hnf}(\rho C_p)_{hnf}^{-1} \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial y} \right) \\
&+ (\rho C_p)_{hnf}^{-1} \frac{4}{3} \frac{T_\infty^3 \sigma^*}{k^*} \frac{\partial}{\partial y} \left(\frac{\partial T}{\partial y} \right) \\
&- \lambda \left[\left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) \frac{\partial T}{\partial y} + v^2 \frac{\partial^2 T}{\partial y^2} + 2uv \frac{\partial^2 T}{\partial x \partial y} + u^2 \frac{\partial^2 T}{\partial x^2} \right] \\
&+ \mu_{hnf}(\rho C_p)_{hnf}^{-1} \left(\frac{\partial u}{\partial y} \right)^2
\end{aligned} \quad (7)$$

$$\begin{aligned}
u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} &= \frac{\partial^2 C}{\partial y^2} D_m - (C - C_\infty) k_0^2 \left(\frac{T}{T_\infty} \right)^n \exp \left(-\frac{E_0}{k_1 T} \right) \\
&\text{at } y = 0 : v = v_w, u = u_w + \gamma^* \left(1 - \zeta^* \frac{\partial u}{\partial y} \right)^{-0.5} \frac{\partial u}{\partial y} \Big|_{y=0}, \\
&u_{yy} = \frac{\partial^2 u}{\partial y^2} = 0, T = T_w, C = C_w, \\
&\text{as } y \rightarrow \infty : u \rightarrow 0, u_y = \frac{\partial u}{\partial y} \rightarrow 0, T \rightarrow T_\infty, C \rightarrow C_\infty.
\end{aligned} \quad (8)$$

For the purpose of converting governing equations, Rao, Thiragarajan, and Kumar (2022) introduced further similarity transformations:

$$\begin{aligned}
\theta &= \frac{T - T_\infty}{T_w - T_\infty}, u = bx \frac{df}{d\eta} = bxf'(\eta), v = -f(bv)^{0.5}, \\
C &= C_\infty + \Phi(\eta)(C_w - C_\infty), \eta = \sqrt{\frac{b}{v}} y.
\end{aligned} \quad (10)$$

Through the use of terms in (10), the continuity equation (5) is satisfied in a straightforward manner. Terms in (10) can then be skilfully used to convert (6-9) as the following:

$$\begin{aligned}
\frac{1}{S_1 S_2} \left(1 + \frac{m-1}{r} (r+1) (We)^r (f'')^r \right) f''' + ff'' - \frac{1}{S_1} Csf' - f'^2 \\
- \frac{S_3}{S_1} Mg f' + \lambda_1 (\theta + \Phi \lambda_1^*) = 0
\end{aligned} \quad (11)$$

$$\frac{(S_4 + Ra)}{S_5 Pr} \theta'' + \left[f\theta' - (ff'\theta' + f^2\theta'') \Lambda + \frac{1}{S_2 S_5} Eck f'^2 \right] = 0 \quad (12)$$

$$\frac{1}{S_C} \Phi'' - \Gamma(1 + \varepsilon\theta) \exp \left(-\frac{Eg}{1 + \varepsilon\theta} \right) \Phi + f\Phi' = 0 \quad (13)$$

$$\begin{aligned}
&\text{at } \eta = 0 : f(\eta) = S, \theta(\eta) = 1, f'''(\eta) = 0, \Phi(\eta) = 1, \\
&f'(\eta) = 1 + \delta f''(\eta)(1 - f''(\eta)\zeta)^{-1/2}, \\
&\text{as } \eta \rightarrow \infty : \theta(\eta) \rightarrow 0, f'(\eta) \rightarrow 0, \Phi(\eta) \rightarrow 0, f'(\eta) \rightarrow 0.
\end{aligned} \quad (14)$$

Here

$$\left. \begin{aligned}
We &= \Upsilon \sqrt{\frac{bu_w^2}{v}}, Pr = \frac{\mu C_p}{k}, Mg = \frac{B_0^2 \sigma}{b\rho}, \Lambda = \lambda b, \\
Ra &= \frac{16}{3} \frac{\sigma^* T_\infty^3}{kk_\infty}, S = -\frac{v_w}{\sqrt{bv}}, \\
Eck &= \frac{u_w^2}{C_p(T_w - T_\infty)}, Sc = \frac{v}{D_m}, \varepsilon = \frac{T_w - T_\infty}{T_\infty}, \Gamma = \frac{k_0^2}{b}, \\
\delta &= \gamma^* \sqrt{\frac{b}{v}}, \zeta = b \sqrt{\frac{b}{v}} \zeta^* x, Eg = \frac{E_0}{k_1 T_\infty}.
\end{aligned} \right\}$$

and

$$\begin{aligned}
S_5 &= \left[(1 - \phi_C)(1 - \phi_T) + \phi_C(1 - \phi_T) \frac{(\rho C_p)_C}{(\rho C_p)_f} \right] + \frac{(\rho C_p)_T}{(\rho C_p)_f} \phi_T, \\
S_{31} &= \frac{\sigma_C - 2\phi_C \sigma_f + 2\phi_C \sigma_C + 2\sigma_f}{\sigma_C + \phi_C \sigma_f - \phi_C \sigma_C + 2\sigma_f}, \\
S_1 &= \phi_T \frac{\rho_T}{\rho_f} + (1 - \phi_T) \left[(1 - \phi_C) + \phi_C \frac{\rho_C}{\rho_f} \right], \\
S_2 &= ((1 - \phi_C)(1 - \phi_T))^{2.5}, \\
S_4 &= S_{41} \times \frac{k_T - 2\phi_T(S_{41}k_f - k_T) + 2S_{41}k_f}{k_T + \phi_T(S_{41}k_f - k_T) + 2S_{41}k_f}, \\
S_3 &= S_{31} \times \frac{\sigma_T + 2S_{31}\sigma_f - 2\phi_T(S_{31}\sigma_f - \sigma_A)}{\sigma_T + 2S_{31}\sigma_f + \phi_T(S_{31}\sigma_f - \sigma_A)}, \\
S_{41} &= \frac{k_C - 2\phi_C(k_f - k_C) + 2k_f}{k_C + \phi_C(k_f - k_C) + 2k_f}.
\end{aligned}$$

Friction factor, Nusselt and Sherwood numbers are outlined as:

$$\begin{aligned}
Cf &= \frac{\mu_{hnf}}{\frac{1}{2}\rho u_w^2} \left(1 + \Upsilon^r \left(\frac{m-1}{r} \right) \left(\frac{\partial u}{\partial y} \right)^r \right) \left(\frac{\partial u}{\partial y} \right) \Big|_{y=0}, \\
Nu &= -x \frac{\left(k_{nf} + \frac{16}{3} \frac{\sigma^* T_\infty^3}{k^*} \right) \frac{\partial T}{\partial y}}{k_f(T_w - T_\infty)} \Big|_{y=0}, \\
Sh &= -x \frac{\frac{\partial C}{\partial y} D_m}{D_m(C_w - C_\infty)} \Big|_{y=0}.
\end{aligned} \quad (15)$$

Terms in (10) allows us to rewrite the terms in (15) as

$$\begin{aligned}
(Re_x)^{0.5} Cf &= \frac{2}{S_2} \left(1 + \frac{m-1}{r} (We)^r (f''(\eta))^r \right) f''(\eta) \Big|_{\eta=0}, \\
(Re_x)^{-0.5} Nu &= -(S_4 + Ra) \theta'(\eta) \Big|_{\eta=0}, \\
(Re_x)^{-0.5} Sh &= -\Phi'(\eta) \Big|_{\eta=0}.
\end{aligned}$$

2.1. Irreversibility formulation

The formula that expresses the dimensional calculation of entropy generation as follows:

$$\begin{aligned}
S_g &= \frac{1}{T_\infty^2} k_f \left(\frac{k_{hnf}}{k_f} + \frac{16}{3} \frac{\sigma^* T_\infty^3}{k_f k^*} \right) \left(\frac{\partial T}{\partial y} \right)^2 + \frac{1}{T_\infty} \mu_{hnf} \left(\frac{\partial u}{\partial y} \right)^2 \\
&+ \frac{1}{T_\infty} \sigma_{hnf} u^2 B_0^2 + \frac{\tilde{R} D_m}{C_\infty} \left(\frac{\partial C}{\partial y} \right)^2 + \frac{\tilde{R} D_m}{T_\infty} \frac{\partial C}{\partial y} \frac{\partial T}{\partial y}.
\end{aligned} \quad (16)$$

By applying (10), equation (16) can be rewritten as follows:

$$EG = (S_4 + Ra) \varepsilon \theta'^2 + \frac{1}{S_2} Br f'^2 + S_3 Mg Br f'^2 + J \frac{\beta_1}{\varepsilon} \Phi'^2 + J \Phi' \theta', \quad (17)$$

Table 2. Validation with former results for $-\theta'(0)$.

	Pr	$-\theta'(0)$
	Ahmed et al. (2024)	Present outcomes
1	1.0000	0.995871
2	1.5227	1.522607
3	1.9236	1.923629
4	2.2607	2.260790
5	2.5575	2.557680

where

$$EG = \frac{v T_{\infty} S_g}{b(T_w - T_{\infty})k_{\infty}}, Br = \frac{\mu u_w^2}{(T_w - T_{\infty})k_f}, \beta_1 = \frac{C_w - C_{\infty}}{C_{\infty}},$$

$$J = \frac{\bar{R}(C_w - C_{\infty})D_m}{k_f} \Bigg\}$$

The Bejan number can be estimated by adopting the subsequent procedure:

$$Be = \frac{\text{Entropy formation due to the transfer of mass and heat}}{\text{The overall generation of entropy}}$$

Be can be obtained in the following way by using (17):

$$Be = \frac{(S_4 + Ra)\alpha\theta'^2 + J\frac{\beta_1}{\varepsilon}\Phi'^2 + J\Phi'\theta'}{(S_4 + Ra)\varepsilon\theta'^2 + \frac{1}{S_2}Brf'^2 + S_3MgBrf'^2 + J\frac{\beta_1}{\varepsilon}\Phi'^2 + J\Phi'\theta'}$$

3. Validation

To validate the reliability of our findings, we perform a comparative study with previously published data acquired under similar constraints. Based on the data shown in Table 2, there exists a substantial degree of consensus.

4. Numerical procedure

The in-built function `bvp4c` in MATLAB is employed to resolve equations (11 - 13) with the constraints (14). This function effectively handles the given conditions (14) to provide accurate solutions. `Bvp4c` implements a finite difference method based on the Lobatto IIIa formula. This method is known for its efficiency in terms of computational cost. Additionally, it's a robust solver, meaning it can handle a wide range of BVPs without encountering stability issues as often compared to some other techniques. Boundary value problems need to be well-posed for any numerical solver to find a unique and stable solution. `bvp4c` won't be able to determine if a problem is ill-posed (missing information or has multiple solutions), and it might struggle to converge in such cases.

Table 3 contains the results of a grid independence test that was conducted to decrease the truncation error. We analyze the grid independence of the present code by evaluating the Nusselt number using various mesh sizes. Based on our analysis, we have concluded that using a grid size of 100 guarantees the implementation of a methodology that is not influenced by the size of the grid. Figure 2 provides the flowchart that explains the numerical technique (`bvp4c` approach).

Table 3. Examination of the Nusselt number at different values and with different grid sizes.

Λ	Mesh size	Nusselt number			
		100	150	200	300
0		1.926793	1.926793	1.926793	1.926793
0.1		2.001308	2.001308	2.001308	2.001308
0.2		2.082285	2.082285	2.082285	2.082285
0.3		2.170553	2.170553	2.170553	2.170553

5. Results with discussion

5.1. Multiple linear regression

One statistical procedure for modelling the connection of a dependent variable with more independent variables is multiple linear regression. It's a powerful tool for predicting outcomes when multiple factors are involved. By analysing the relationships between these variables, we can understand how changes in one variable affect the other. This technique has a huge usage in various fields, including economics, social sciences and engineering, to make data-driven decisions and gain insights into complex phenomena.

Crucial technical characteristics, such as the friction factor and specific variables like Weissenberg number were linked in this study using the models outlined below:

$$Cf = p_0 + p_1We + p_2Mg, \quad (18)$$

$$Nu = q_0 + q_1Eck + q_2\Lambda, \quad (19)$$

$$Sh = r_0 + r_1Eg + r_2\Gamma, \quad (20)$$

We used 20 different sets of variables for each computation to speed up the process of discovering results:

$$Cf = -3.5871 + 3.6661We - 0.6122Mg, \quad (21)$$

$$Nu = 1.9718 - 3.0611Eck + 0.3862\Lambda, \quad (22)$$

$$Sh = 0.3957 - 0.0346Eg + 0.1975\Gamma, \quad (23)$$

According to equation (21), the friction coefficient declines as Mg surges, but it upsurges as We surges (see Figure 3 and Table 4). When the Weissenberg number increases, its viscosity decreases with increasing shear rate. As the fluid flows over a surface, the shear rate near the surface is higher, leading to a decrease in viscosity. This reduction in viscosity, however, can result in increased shear stress at the surface, causes a higher skin friction coefficient. The friction coefficient rises by 34.4% as the value of We ranges from 0 to 0.5. In industries like food processing or chemical manufacturing, where non-Newtonian fluids are common, understanding the relationship between fluid properties and skin friction is crucial. For instance, enhancing the Weissenberg number (Carreau-Yasuda parameter) in a shear-thinning fluid could lead to higher energy consumption in pumps or mixers due to increased frictional losses. The Nusselt number declines when Eckert number (Eck) increases, as shown in Equation (22), and the opposite is true when thermal relaxation parameter (Λ) surges (see Figure 4 and Table 5). A higher Λ implies a slower response of the heat flux to changes in the temperature gradient. This results in a thicker thermal boundary layer, as the heat takes longer to propagate through the fluid. Despite the thicker boundary layer, the temperature

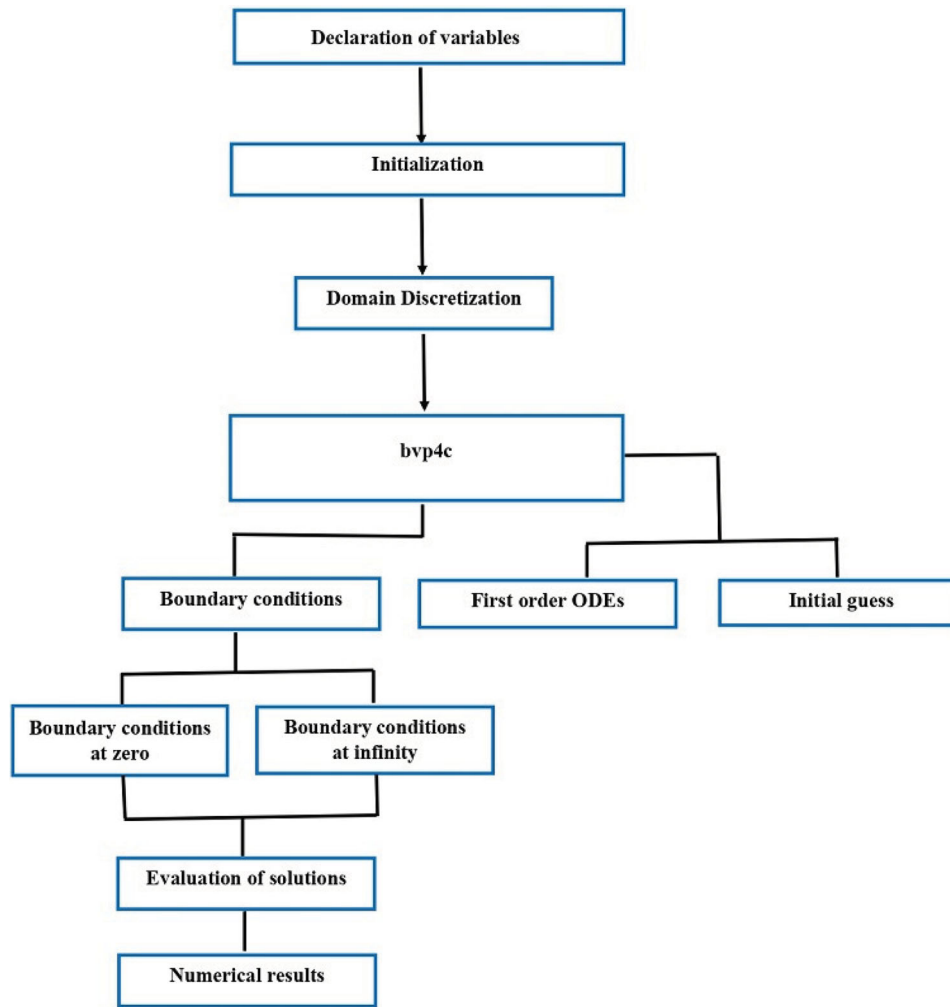


Figure 2. The bvp4c framework (Hussain et al. 2022).

gradient near the surface becomes steeper because the heat flux is delayed. The steeper temperature gradient enhances the heat transfer rate at the surface, leading to a higher Nusselt number. An upsurge in the Eckert number indicates a higher proportion of kinetic energy transmuted into thermal energy through viscous dissipation, leading to an upsurge in the fluid's temperature, reducing the temperature difference between the fluid and the surface. Further, the thermal gradient near the surface lessens, leading to a minimised thermal conductivity. Enhancing the thermal relaxation parameter effectively delays the fluid's response to temperature changes, causing a thicker thermal boundary layer and a steeper temperature gradient near the surface. This ultimately results in enhanced heat transfer, as reflected by a higher Nusselt number. Research has revealed that the Nusselt number experiences a rise of 8.8% when the thermal relaxation parameter is between the range of 0–0.4. In processes like chemical reactions or heat exchangers, controlling the thermal relaxation parameter can be utilised to regulate the rate of heat transfer, affecting factors like reaction speed or energy efficiency. It is evident from equation (23) that there is an escalation in the Sherwood number with a boost in Υ but it drops with an escalation in E_g (see Figure 5 and Table 6). A surge in Υ causes a higher Sherwood number due to the enhanced consumption of the species at the surface. This increased consumption creates a

steeper concentration gradient near the surface, driving faster diffusion and consequently a higher mass transfer rate, which is directly reflected in the Sherwood number. Augmenting activation energy parameter causes a decrement in reaction rate. With a slower reaction, the concentration gradient near the surface becomes less steep. Consequently, the Sherwood number decreases. Furthermore, the Sherwood number diminishes at a rate of 3.2% when E_g lies between 0 and 3. In bioreactors used for cell culture or fermentation, an increase in activation energy for the metabolic processes of the cells can reduce the rate of mass transfer of nutrients and products between the cells and the surrounding fluid. This has the potential to affect the bioreactor's overall production.

5.2. Impact of various parameters on velocity profile

An increase in Mg typically contributes a decay in the velocity profile. This outcome occurs because of the magnetic field employing a driving force on the fluid, therefore counteracting its movement. The magnitude of the opposing force intensifies as the magnetic field intensity upsurges, resulting in the deceleration of the fluid (see Figure 6). In real-world scenarios, this principle is utilised in various fields, for instance, in metallurgy,

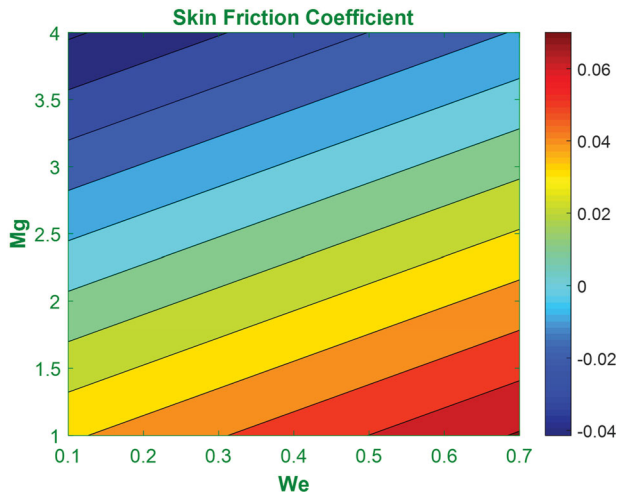


Figure 3. Situation where Mg and We influences Cf .

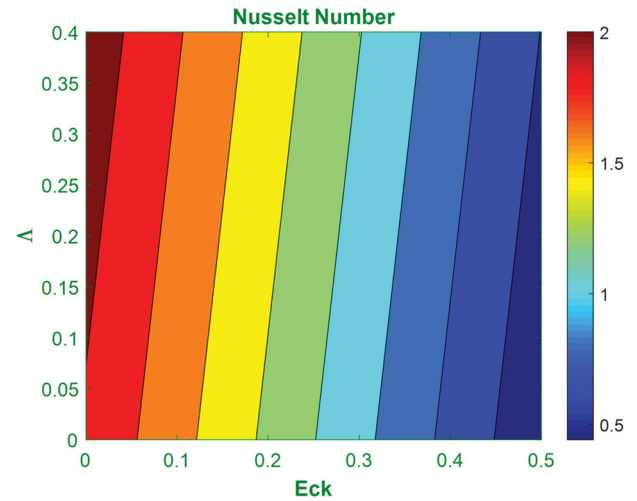


Figure 4. Situation where Λ and Eck influences Nu .

during the casting process, a magnetic field is supplied to command the flow of molten metal, preventing defects and ensuring uniform solidification. An enhancement in Cs causes an associated rise in velocity of the fluid (see Figure 7). This phenomenon occurs due to the inclusion of the internal rotating motion of fluid particles in the couple stress component. A superior value of this characteristic recommends that the fluid particles possess a higher degree of rotational freedom. The augmented rotating motion can serve to decrease fluid resistance, therefore enabling the attainment of greater flow velocities. In essence, the fluid experiences a decrease in viscosity, therefore facilitating its flow. In nanoscale systems, the couple stress parameter plays a significant role in identifying the fluid behaviour. Increasing this parameter can enhance the transport of fluids through nanostructures, which has applications in fields like nano fluidics and nanomaterials. An increment in the volume fraction of nanofluids causes a concurrent drop in the fluid's velocity (see Figure 8). The effective viscosity of a nanofluid get larger proportionately with the volume percentage of nanoparticles present in the fluid. The basis for this occurrence is that the nanoparticles function as impediments to the flow, therefore magnifying the resistance to fluid motion. The Weissenberg number (Carreau-Yasuda fluid parameter) quantifies the fluid's shear-thinning or shear-thickening behaviour. When this parameter increases, it signifies a greater deviation from Newtonian fluid behaviour. In shear-thinning fluids, as the shear rate get larger, the fluid's viscosity lessens, causes to a reduction in resistance to flow and consequently, an increase in velocity. Conversely, in shear-thickening fluids, the opposite occurs: increasing the shear rate leads to an increase in viscosity, causes a greater resistance to flow and a decrease in velocity. Therefore, an upsurge in We generally causes a plunge in the velocity of the fluid. (see Figure 9).

5.3. Influence of different factors on fluid temperature

Lower fluid temperatures are produced by higher thermal relaxation time parameter values (see Figure 10). The thermal relaxation time is the duration needed for a fluid particle to adapt

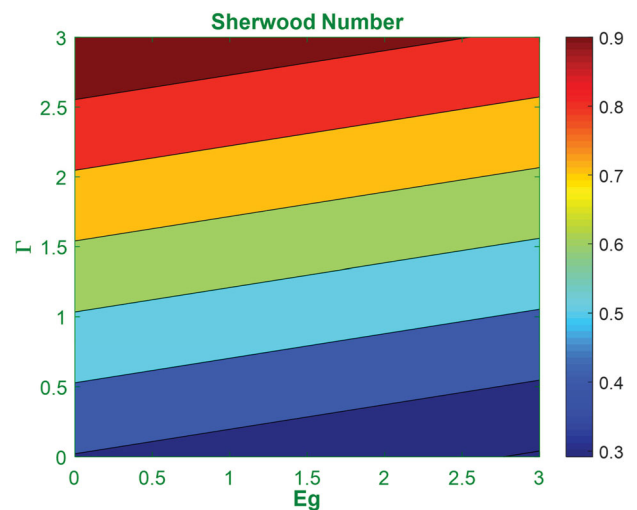


Figure 5. Situation where Γ and Eg influences Sh .

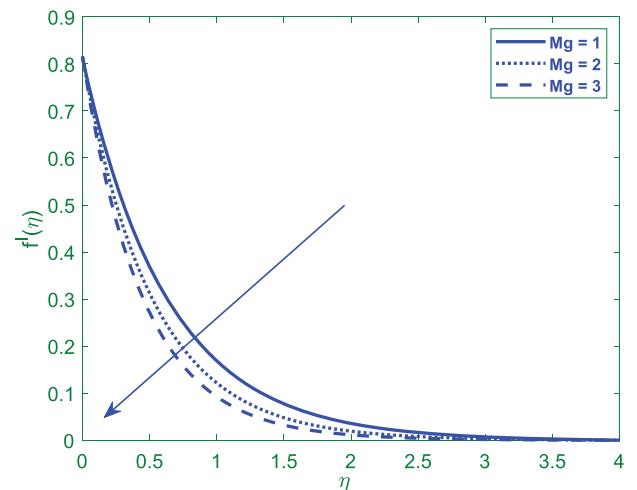


Figure 6. Situation where Mg influences $f'(\eta)$.

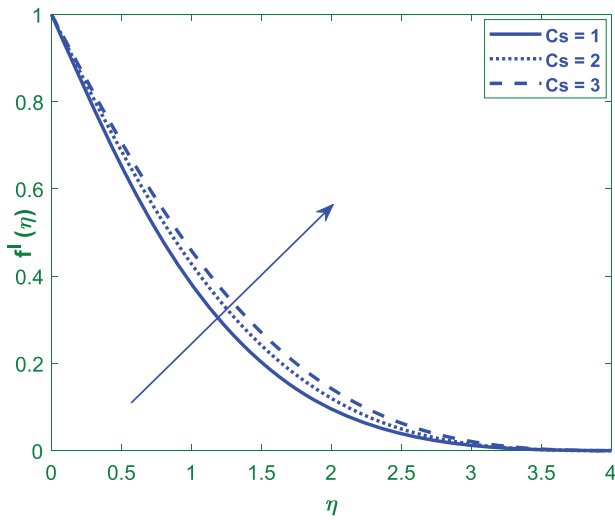


Figure 7. Situation where C_s influences $f'(\eta)$.

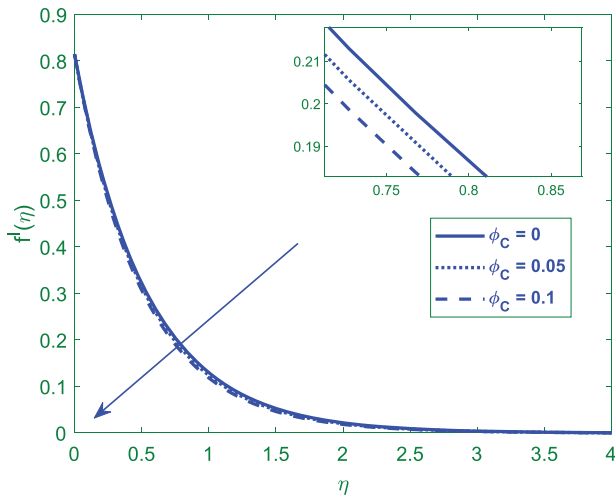


Figure 8. Situation where ϕ_c influences $f'(\eta)$.

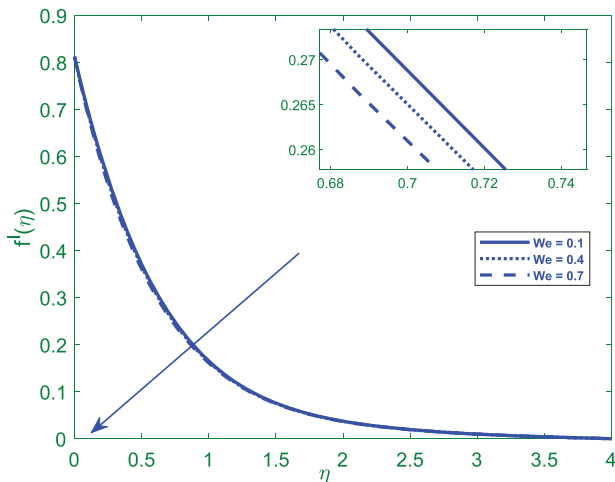


Figure 9. Situation where We influences $f'(\eta)$.

Table 4. Influence of Mg and We on C_f .

Mg	We	Skin friction coefficient
1		−3.09838
2		−3.80294
3		−4.38505
4		−4.88828
	0.1	−3.84306
	0.3	−3.80294
	0.5	−3.70342
	0.7	−0.88281

Table 5. Influence of Eck and Δ on Nu .

Eck	Δ	Nusselt number
0		2.26082
0.1		1.96533
0.2		1.67080
0.3		1.37725
	0	0.84401
	0.15	0.85595
	0.225	0.86253
	0.3	0.86955

Table 6. Influence of Eg and Γ on Sh .

Eg	Γ	Sherwood number
0		0.465524
1		0.405508
2		0.378052
3		0.365969
	0	0.356748
	1	0.608884
	2	0.795886
	3	0.948519

its temperature to the pressure of its surroundings. As the duration of this time grows, the fluid particles need more time to transmit heat to adjacent particles. In addition, the rate of heat transmission decreases, resulting in a reduced temperature. In microelectronics cooling systems, where heat dissipation is a critical concern, reducing the thermal relaxation time can help prevent overheating and improve device performance. The growth of thermal radiation leads to a temperature escalation of a fluid, since thermal radiation conveys energy via electromagnetic waves from a hotter surface or environment to the fluid. When the fluid bolt up this radiation, the energy stimulates its molecules, elevating their kinetic energy. This molecular agitation becomes the reason for an enhancement in the fluid's temperature (see Figure 11). In solar collectors and thermal power plants, the increased thermal radiation from the sun can significantly boost the temperature of working fluid, improving efficiency. An escalation in the Eckert number indicates a relatively high kinetic energy of the fluid about its internal energy. The greater kinetic energy results in a higher frequency of collisions among fluid particles, therefore producing heat by friction. This phenomenon is referred to as viscous dissipation. The overall temperature of the fluid upsurges due to the transformation of kinetic energy into internal energy, which is expressed as heat (see Figure 12).

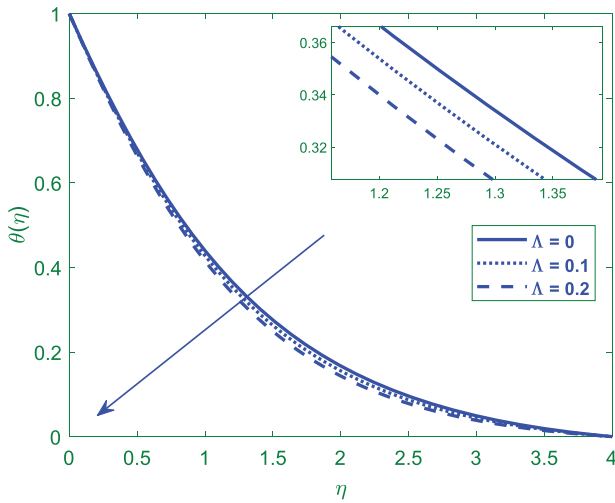


Figure 10. Situation where Λ influences $\theta(\eta)$.

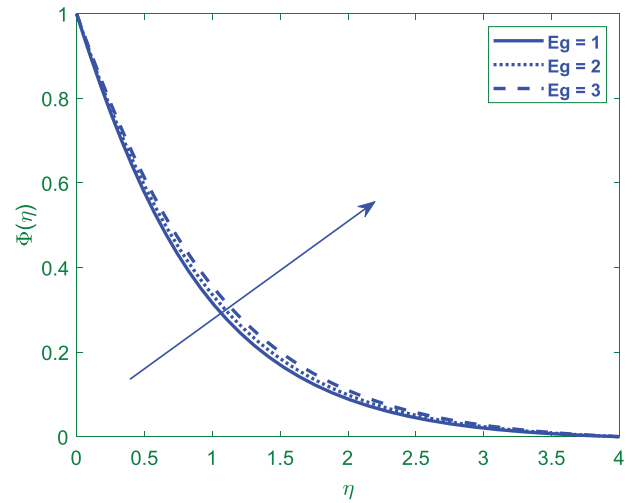


Figure 13. Situation where E_g influences $\Phi(\eta)$.

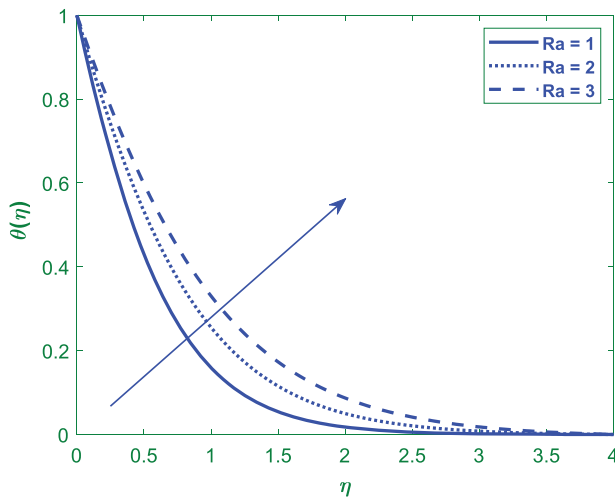


Figure 11. Situation where Ra influences $\theta(\eta)$.

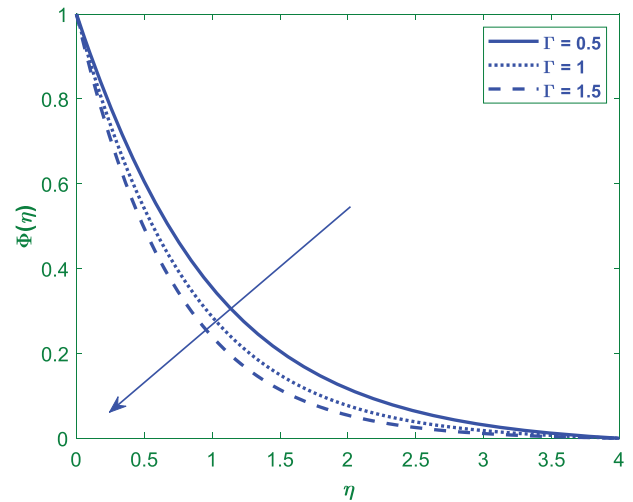


Figure 14. Situation where Γ influences $\Phi(\eta)$.

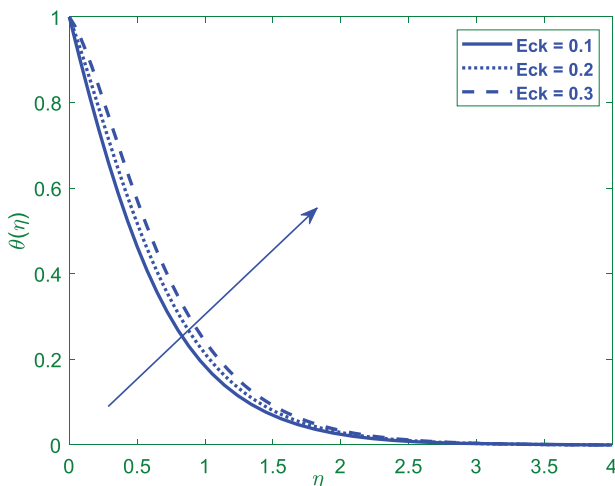


Figure 12. Situation where Eck influences $\theta(\eta)$.

5.4. Impact of various parameters on concentration profile

Heightened activation energy results in a corresponding rise in concentration of the fluid (see Figure 13). Figure 14 illustrated that the surge in Γ diminishes the concentration of the fluid. The rationale for this is that a greater reaction rate parameter signifies a more rapid occurrence of the chemical reaction. By increasing the rate of the reaction, the reactants are depleted at an accelerated pace, resulting in a reduction in their concentration. Reactants in the context of a fluid refer to the constituent chemical elements that constitute the fluid. Because of this reason, outcome in Figure 14 occurs.

5.5. Irreversibility analysis

Huge values of Brinkman number result in greater entropy production in fluid flow (see Figure 15). Increment in the Brinkman number suggests that the influence of viscous dissipation is growing in comparison to heat conduction. The enhanced dissipation rate is reason for the production of additional heat inside

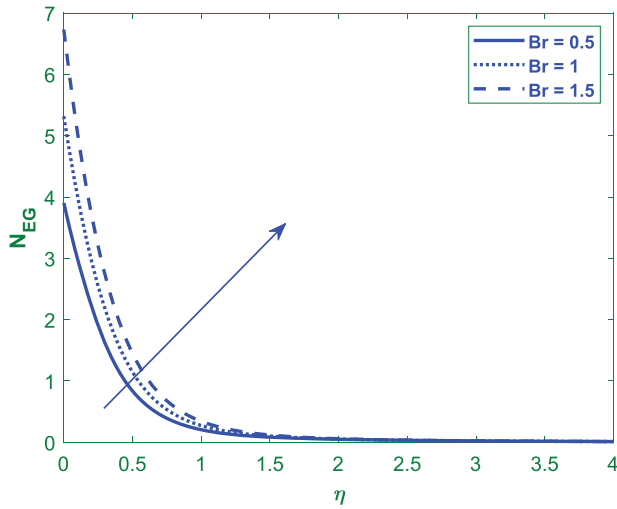


Figure 15. Situation where Br influences N_{EG} .

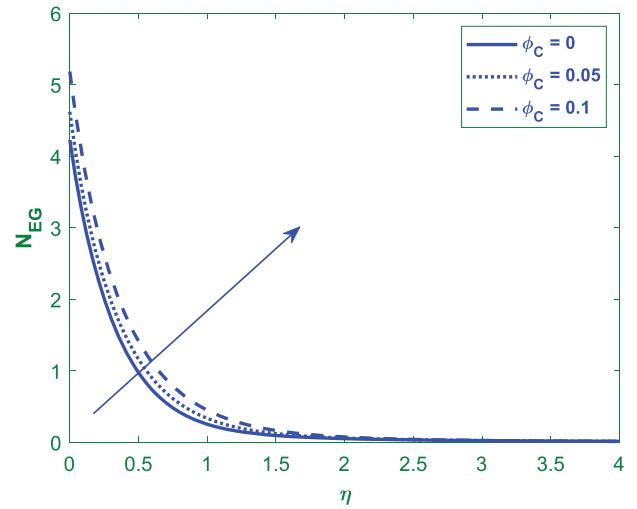


Figure 16. Situation where ϕ_C influences N_{EG} .

the fluid and thus causing a greater level of disorder and an augmentation in entropy. A rise in ϕ_C typically results in a relative rise in the entropy production (see Figure 16). The main cause of this phenomenon is the increased thermal conductivity and viscosity of the nanofluid in comparison to the base fluid. Enhancing the volume fraction of nanoparticles causes an improvement in the thermal conductivity of the fluid, thereby facilitating more effective heat transfer. Nevertheless, the enhanced heat transfer can also lead to an elevated rate of entropy production, since heat transfer is frequently accompanied by a rise in disorder. A real-world application of the increased entropy generation caused by rising nanoparticle volume fraction in fluid flow is in microelectronics cooling. As the demand for smaller and more powerful electronic devices increases, efficient heat dissipation becomes crucial to prevent overheating and component failure. Nanofluids, with their enhanced thermal conductivity, are promising candidates for cooling these devices. However, increasing the nanoparticle volume fraction in these fluids can lead to a trade-off: while improving heat transfer, it also increases entropy generation, which can reduce efficiency of the overall system. Heightening the Brinkman number leads to a reduction in the Bejan number. Also, the enhancement in the Brinkman number signifies an increase in entropy production associated with fluid friction. The rise in entropy production by fluid friction helps to an escalation in the overall irreversibility of the system, thereby causing a decrement in the Bejan number (see Figure 17). A growth in the percentage of nanoparticles in the volume of a fluid flow results to a decrement in the Bejan number (see Figure 18). It is observed that the thermal conductivity and viscosity of the fluid exhibit a positive correlation with the volume fraction of nanoparticles. This increased thermal conductivity facilitates the transfer of heat, therefore decreasing the irreversibility common with heat transfer. Nevertheless, the augmented viscosity results in elevated fluid friction, therefore contributing to a rise in the overall irreversibility. Given that the rise in heat transfer irreversibility is typically less substantial than the rise in fluid friction irreversibility, the overall outcome is a reduction in the Bejan number.

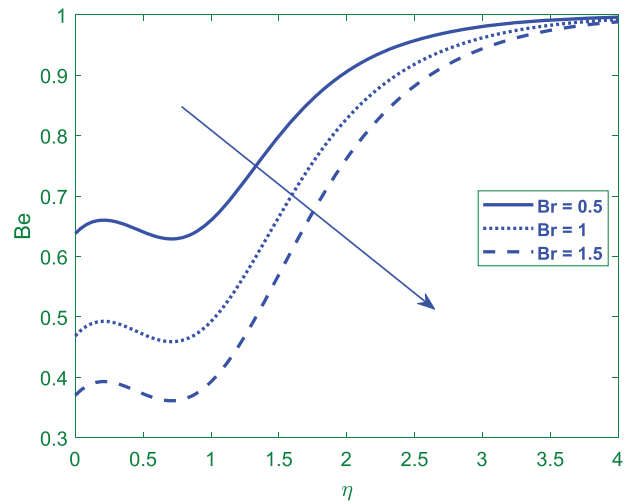


Figure 17. Situation where Br influences Be .

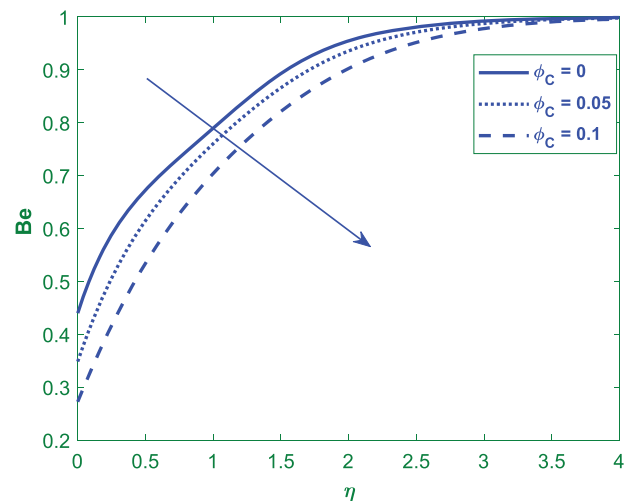


Figure 18. Situation where ϕ_C influences Be .

6. Conclusions

The research aimed to examine the scenarios of flow and heat transmission in the Carreau-Yasuda hybrid nanofluid moving across an elastic sheet. The governing equations were formulated under the assumptions of steady-state, incompressible flow, and boundary layer approximations. The equations were subsequently simplified and numerically addressed via the bvp4c technique within MATLAB. Key findings mentioned here:

- A rise in both the Brinkman number and volume percentage of nanoparticles causes an escalation in the entropy production. This means that the system becomes less efficient, as more energy is dissipated as heat rather than being utilised for useful work. By carefully controlling the Brinkman number and nanoparticle volume fraction, engineers can minimise entropy generation and improve the overall efficiency of these systems.
- Higher values of the Weissenberg number led to increased skin friction coefficient and reduced velocity. In processes like extrusion and injection moulding, increased Weissenberg numbers can lead to higher energy consumption and potential equipment damage due to increased pressure drops.
- The magnetic field parameter had a notable influence on the velocity profile, causing it to decrease with increasing magnetic field strength. This is due to the Lorentz force opposing the fluid motion. This phenomenon has practical applications in various fields, for instance, magnetic fields are used to control the flow of molten metals during casting processes.
- The thermal relaxation parameter, which accounts for non-Fourier heat flux, influenced the temperature distribution. Higher values of this parameter led to a thicker thermal boundary layer and reduced temperature gradients. This phenomenon is relevant in various fields, including materials science, heat transfer, and fluid dynamics. Understanding the thermal relaxation parameter is crucial for designing materials and systems with specific thermal properties.
- An upsurge in the nanoparticles' volume fraction causes to a reduction in the Bejan number.
- The Nusselt number was enhanced by 8.8% due to the increment in Δ .
- Heightened activation energy results in a corresponding rise in fluid concentration
- The mass transmission number decreased by 3.2% when the activation energy parameter increased from 0 to 3.

Future scope of the work

Here are some key areas of future research:

- Investigate flows over curved surfaces, cylindrical surfaces, or spheres to mimic real-world applications like heat exchangers and microfluidics.
- Study the flow and heat transfer of Carreau-Yasuda hybrid nanofluids through porous media, relevant to applications like geothermal energy and filtration processes.
- Design and conduct experiments to validate the theoretical predictions and numerical simulations.

- Utilise machine learning algorithms to predict the behaviour of nanofluids and optimise design parameters.

Author contributions

CRediT: **S. Suneetha**: Conceptualization; **Rajavath Narayana Naik**: Supervision;

Disclosure statement

No potential conflict of interest was reported by the author(s).

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